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|---|---|
| Date | April 27th, 2026 |
| Group | 322 |
| Member | Tomas Chavarria |
| Member | |
| Homework | Project |
| Due | April 28th, 2026 |

1. Project Description:
   This project is meant to implement a database management system which houses information on a rental service. More specifically, it covers the current rentals, items, customers, and returns. It allows for filtering of any given category and for various checks on different data, such as (but not limited to): filtering items or customers, waitlists for items, rental extensions, etc. We are supposed to implement the underlying SQL functionality, with the database creation and main loop for actually inserting being provided to us.

2. What works:
   - add_customer
   - add_item
   - edit_customer
   - rent_item
   - waitlist_customer
   - update_waitlist
   - return_item
   - grant_extension
   - get_filtered_items
   - get_filtered_customers
   - get_filtered_rentals
   - get_filtered_rental_histories
   - get_filtered_waitlist
   - number_in_stock
   - place_in_line
   - line_length
   - save_changes
   - close_connection

3. What does not work:
   - N/A

4. Known Bugs:
   - When running the tests from public_tests.py as batch in Visual Studio Code, many of the tests will fail or pass sporadically. When checking the tests individually, the only test that fails about 30% of the time is close_connection. The reason for this is unknown.

Tests were not run in PyCharm. When manually testing the program, no notable bugs were found.

5. Required Screenshots:

```
PS D:\IMPORTANT-INFORMATION\University of Florida\Junior Spring\Databases\Final Project\cis4301sp26_project> python setup_db.
py

Using:
        Username: root
        Password:
        Port: 3306
        Data Directory: tpcds_data

Connected to the DB
Creating tables...
Tables created successfully
Loading item.csv...
  Loaded 18000 rows into item
Loading customer_address.csv...
  Loaded 50000 rows into customer_address
Loading customer.csv...
  Loaded 100000 rows into customer
Loading date_dim.csv...
  Loaded 73049 rows into date_dim
Loading store_sales.csv (this may take several minutes)...
  Loaded 2880404 rows into store_sales
Setting i_num_owned to random values (1-5)...
  Updated 18000 items
Populating rental_history from store_sales...
  Inserted 2685933 rows into rental_history

Database setup complete!
Successfully loaded in the data
```

1. Load setup_db.py

```
PS D:\IMPORTANT-INFORMATION\University of Florida\Junior Spring\Databases\Final Project\cis4301sp26_project> python main.py
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: AAAAAAAABAAAAAA
Enter Customer ID: AAAAAAAABAAAAAA
Successfully rented item
```

2. Rent an item

3. Return an item

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 2

Enter Item ID: AAAAAAAABAAAAAA
Enter Customer ID: AAAAAAAABAAAAAA
Successfully returned the item
```

4. Grant an extension

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 3

Enter Item ID: AAAAAAAABAAAAAA
Enter Customer ID: AAAAAAAABAAAAAA
Successfully granted extension

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 3

Enter Item ID: AAAAAAAABAAAAAA
Enter Customer ID: AAAAAAAABAAAAAA
The customer already has an extension and may not be granted another one
```

5. Filtered item search using at least two filters

```
Current Filters:
--------------------
Category: Electronics
Min Start Year: 2000
--------------------

Which attribute would you like to filter?
1. Item ID
2. Product Name
3. Brand
4. Manufacturer
5. Category
6. Min Price
7. Max Price
8. Min Start Year
9. Max Start Year
10. Continue
11. Cancel

Choice: 10
```

```
--------------------
Item ID: AAAAAAAAOBGEAAAA
Product Name: ableantin stationought
Brand: exportiamalgamalg #1
Category: Electronics
Manufacturer: esebarable
Current Price: 1.73
Start Year: 2001
Total number of copies owned: 2
--------------------
--------------------
Item ID: AAAAAAAAMDGEAAAA
Product Name: ableeingn stationought
Brand: maxiunivamalg #14
Category: Electronics
Manufacturer: barbarought
Current Price: 9.29
Start Year: 2001
Total number of copies owned: 5
--------------------

Found 618 items.
```

6. Customer search using pattern

```
Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 2

Would you like to use patterns to search String attributes? (Y/N): Y
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 2

Name: %son

Current Filters:
--------------------
Name: %son
--------------------
```

```
--------------------
Customer ID: AAAAAAAACGGIBAAA
Name: Stephen Olson
Address: 939 5th 2nd, Lebanon, NC 22898
Email: Stephen.Olson@B76AUL5jrh.com
--------------------
--------------------
Customer ID: AAAAAAAAGGGIBAAA
Name: Dennis Anderson
Address: 383 6th Johnson, New Hope, WI 59431
Email: Dennis.Anderson@dPnKhA8OE2.com
--------------------
--------------------
Customer ID: AAAAAAAAJHGIBAAA
Name: Thomas Peterson
Address: 495 8th Sycamore, Glendale, NC 23951
Email: Thomas.Peterson@VI9CkEbcLdCSS1dCR.com
--------------------

Found 7632 customers.
```

7. Rental search filtered by date range

```
Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Continue
8. Cancel

Choice: 7


Current Filters:
--------------------
Min Rental Date: 2026-04-20
Max Rental Date: 2026-05-20
--------------------

--------------------
Item ID: AAAAAAAABAAAAAAA
Customer ID: AAAAAAAABAAAAAAA
Rental Date: 2026-04-26
Due Date: 2026-05-24
--------------------

Found 1 rental.
```

8. Rental history search showing results populated from store_sales (filtered by date range to minimize entries)

```
Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 4

Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Rental Date
4. Max Rental Date
5. Min Due Date
6. Max Due Date
7. Min Return Date
8. Max Return Date
9. Continue
10. Cancel

Choice: 9
```

```
--------------------
Item ID: AAAAAAAAPPMBAAAA
Customer ID: AAAAAAAACCHHBAAA
Rental Date: 2000-04-05
Due Date: 2000-04-19
Return Date: 2000-04-19
--------------------
--------------------
Item ID: AAAAAAAAPPNAAAAA
Customer ID: AAAAAAAACNHABAAA
Rental Date: 2000-04-03
Due Date: 2000-04-17
Return Date: 2000-04-17
--------------------

Found 4493 returns.
```

9. Add customer to waitlist, search waitlist

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 1

Enter Item ID: AAAAAAAABAAAAAAA
Enter Customer ID: AAAAAAAAEAAAAAAA
This item is not available right now.
Would you like to waitlist the Customer (Y/N): Y
The customer is now 1st in line to rent the item
```

```
Which attribute would you like to filter?
1. Item ID
2. Customer ID
3. Min Place in line
4. Max Place in line
5. Continue
6. Cancel

Choice: 5


Current Filters:
-------------------
Item ID: AAAAAAAABAAAAAAA
-------------------

-------------------
Item ID: AAAAAAAABAAAAAAA
Customer ID: AAAAAAAAEAAAAAAA
Place in line: 1
-------------------

Found 1 waitlisted customer.
```

10. Add a new item, then duplicate ID rejection

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 5

Enter Item ID: 1
Enter Product Name: newprod
Enter Brand: NewProd
Enter Category: Electronics
Enter Manufacturer: NewProd
Enter Current Price: 5.00
Enter Start Year: 2020
Enter the number of copies owned: 1

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 5

Enter Item ID: 1
An item with that ID already exists.
```

11. Add customer, then duplicate ID rejection

```
What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 6

Enter Customer ID: 1
Enter First Name: Joe
Enter Last Name: Hart
Enter Email: joe.hart@gmail.com
Enter Street Number: 121 Main St
Enter Street Name: Main
Enter City: Miami
Enter State: FL
Enter Zip: 33149

What would you like to do?
1. Rent an Item
2. Return an Item
3. Grant an Extension
4. Search a Table
5. Add an Item
6. Add a Customer
7. Edit a Customer
8. Exit

Choice: 6

Enter Customer ID: 1
A customer with that ID already exists.
```

12. Edit two fields then search (note the previous address was 121 Main St Main, here changes to 121 Main St.)

```
Customer's Customer ID: 1

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 3

Address: 121 Main St.
Current Attributes:
--------------------
Address: 121 Main St.
--------------------

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 4

Email: joe.hart@yahoo.com
Current Attributes:
--------------------
Address: 121 Main St.
Email: joe.hart@yahoo.com
--------------------

Which attribute would you like to edit?
1. Customer ID
2. Name
3. Address
4. Email
5. Save
6. Cancel

Choice: 5

Current Attributes:
--------------------
Address: 121 Main St.
Email: joe.hart@yahoo.com
--------------------
```

```
Which table would you like to search?
1. Item
2. Customer
3. Rental
4. Rental History
5. Waitlist
6. Cancel

Choice: 2

Would you like to use patterns to search String attributes? (Y/N): N
Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 1

Customer ID: 1

Current Filters:
--------------------
Customer ID: 1
--------------------

Which attribute would you like to filter?
1. Customer ID
2. Name
3. Address
4. Email
5. Continue
6. Cancel

Choice: 5


Current Filters:
--------------------
Customer ID: 1
--------------------

--------------------
Customer ID: 1
Name: Joe Hart
Address: 121 Main St.
Email: joe.hart@yahoo.com
--------------------

Found 1 customer.
```